High-rank symmetric markings from finite cores

Human Compression project

Complete source recovery; author edition awaiting independent review

Abstract

We recover the principal classification of Fernandes and Leemans for C-groups of ranks $n-1$ and $n-2$ in degree n, with the original bounds $n \geq 7$ and $n \geq 9$. The common theory supplies a uniform intransitive-parabolic reduction and a finite-core attachment theorem. We prove the complete fracture-graph converse, including the local splitting obstructions, and realize every flag by a decorated four-point core and a laminar tree partition. This gives all the asserted regular hypertopes. Exact changes of marking give presentations for arbitrary branching. One clause of the accepted manuscript's printed presentation is incompatible with such branching; we exhibit the literal obstruction and give the corrected general rule. No conjecture solution or new historical classification is claimed.

We use the C-group convention of the companion common paper [U]. The subgroup G_i denotes the maximal parabolic obtained by omitting i; other generated subgroups are written with their full generating sets. The permutation graph has one labelled edge for each transposition factor of a distinguished involution. Its graph may have parallel edges with different labels. An action on cosets will be represented by images of its explicitly described stabilized objects, so left cosets are used throughout.

The source is the accepted 2018 institutional manuscript [FL], with 20 authored pages and a publisher cover. Its five principal targets are Theorems 1.1 and 1.2, Proposition 3.1, Corollary 6.5 and Theorem 6.6. The first, second, third and fifth conclusions are recovered in their full stated ranges. The general presentation conclusion is recovered by an exact corrected rule; its false branching subclause is not endorsed. The ledger records auxiliary statements and exclusions separately from this principal contract.

1 Maximum rank: one transposition for each tree edge

Theorem 1. *Let $n \geq 7$ and let $G \leq S_n$ have a C-group marking of rank $n-1$. Then $G = S_n$, and its distinguished generators are exactly the transpositions on the edges of a tree on the n letters. Conversely every such tree gives a C-group marking of rank $n-1$. Its Coxeter diagram is the line graph of the tree, and its coset system is a regular hypertope with full type-preserving group S_n.*

Proof. The marking is an independent family of length $n-1$, so the equality clause of Whiston's theorem gives $G = S_n$. Retain the omission alternatives recorded in [CC, proof of Theorem 2.1]: each maximal standard parabolic is intransitive, transitive imprimitive with blocks of size two, or A_n. This is an explicit outside input; we do not infer it from the rank-$n-2$ source result being recovered.

In the imprimitive case write $n = 2m$, so $m \geq 4$. The binary-module proof in [U] gives an irredundant-length bound $2m-4$ if the block image is a proper subgroup of S_m, and $m+1$ if it is S_m. Both are strictly smaller than $2m-2$, even for $m = 4$. But our parabolic has $n-2 = 2m-2$ independent generators. The A_n case is excluded by the elementary intersection argument in [U]: the omitted involution must centralize every other generator, whereas the centralizer of A_n in S_n is trivial. Hence every maximal parabolic G_i is intransitive.

For every i, choose an i-edge joining two distinct G_i-orbits in the permutation graph. These $n-1$ edges form a forest: in a cycle the selected i-edge could be bypassed using other labels, contradicting

its orbit choice. Thus they form a spanning tree T. An extra i-edge would, with its path in T, form a cycle. Every selected edge on that path must have label i, by the same bypass argument. Since only one selected i-edge exists, that path is the selected edge itself, not a new edge. There are no extra edges, and all distinguished permutations are the tree transpositions.

Conversely the transpositions on a connected tree generate S_n. For any edge subset F, they generate the product of symmetric groups on the connected components of $(V(T), F)$. Two vertices are in a common component of both F and E exactly when their unique tree path lies in $F \cap E$. Intersecting the two Young subgroups therefore gives precisely the subgroup for $F \cap E$, proving the full intersection property. Two different edge transpositions have product order three when their edges meet and order two otherwise, giving the asserted line graph.

For the geometric claim root T at any vertex. An object of the type corresponding to an edge e is a subset of size equal to the component of $T - e$ away from the root, because its maximal parabolic is the stabilizer of that subset. Two such objects are incident precisely when they have the prescribed disjointness or containment. For a flag of these subsets, subtract the maximal selected children from each selected set and take the complement of the maximal selected sets for the root region. The resulting disjoint regions have exactly the cardinalities of the corresponding standard regions of T. Independent bijections on the regions combine to a permutation realizing the whole flag. Thus the system is flag-transitive. The elementary coset argument proved in Section 10 gives thinness, residual connectedness and regularity, including the full automorphism group. □

The uniform tree-presentation theorem of [U] now recovers [FL, Proposition 3.1]: use the Coxeter relations of the line graph and the triangle relations $(s_e s_f s_e s_g)^2 = 1$. That proof is valid for every finite tree, even when $n < 7$; the classification theorem above retains its stated range $n \geq 7$.

2 The three four-letter cores

Take $Q = \{1, 2, 3, 4\}$ and $I = \{0, 1\}$. The three rooted cores are

$$A: \quad h_0 = (2\ 3), \quad h_1 = (1\ 2)(3\ 4), \quad d = 4;$$

$$B: \quad h_0 = (2\ 3), \quad h_1 = (1\ 2)(3\ 4), \quad d = 3;$$

$$C: \quad h_0 = (1\ 2)(3\ 4), \quad h_1 = (1\ 3)(2\ 4), \quad d = 4.$$

In each case let π_S be the orbit partition of H_S on Q. For the empty set it is discrete and for I it has one block. The two singleton partitions are respectively

$$A, B: \quad 1 \mid 23 \mid 4, \qquad 12 \mid 34; \qquad C: \quad 12 \mid 34, \qquad 13 \mid 24.$$

Their common refinement is discrete, so the partition-meet hypothesis of the attachment theorem in [U] holds.

In A and B the full core is the dihedral group of order eight preserving the unordered pair $\{\{1, 4\}, \{2, 3\}\}$. Its intersections with the two displayed Young subgroups are $\langle h_0 \rangle$ and $\langle h_1 \rangle$ respectively. For the second assertion, among the four permutations independently exchanging $1, 2$ and $3, 4$, just the identity and their simultaneous exchange preserve that unordered pair. In C the full core is $\{1, (12)(34), (13)(24), (14)(23)\}$, and its intersections with the two displayed Young subgroups are again the indicated cyclic groups. The singleton groups intersect trivially, and each nonidentity generator fails to preserve the other singleton partition. This proves the core and mixed subgroup identities in [U], including the empty and full sets.

For saturation, the empty-set case is immediate. A singleton generator which is a transposition either joins d to another core letter or is disjoint from d; adjoining $(d\ x)$ yields exactly the appropriate Young group. If the generator is a double transposition $h = (u\ v)(d\ w)$, then

$$\big(h(d\ x)\big)^3 = (u\ v),$$

so we obtain both $(u\ v)$ and $(d\ w)$, as well as $(d\ x)$. They generate $\mathrm{Sym}(\{u,v\}) \times \mathrm{Sym}(\{d,w,x\})$. Finally H_I is transitive on Q. Conjugating $(d\ x)$ by its elements gives $(q\ x)$ for every $q \in Q$, which generate $\mathrm{Sym}(Q \cup \{x\})$. Thus saturation holds in every case.

Corollary 2. *Attach any finite tree to one of these cores at its displayed root, with that root a leaf of the attached tree. The displayed two involutions and all its edge transpositions form a C-group marking of* $\mathrm{Sym}(\Omega)$ *of rank* $|\Omega| - 2$.

Proof. The finite hypotheses were proved above, and the attached tree has $|\Omega| - 4$ edges. Apply the core-attachment theorem of [U]. □

These are precisely the diagrams in [FL, Theorem 1.2], verified against the accepted PDF and the authors' original TeX. Family A has the path $1 -^1 2 -^0 3 -^1 4$; family B attaches at vertex 3 instead of 4, and C is the alternating square. The tail has one first edge at the root, but may branch at its next vertex. The proof gives all positive intersection assertions at once. The following sections prove exhaustion and the complete geometric claim.

3 Why the leaf cannot be dropped

In core A, attach two new vertices 5 and 6 directly to 4. The marking subsets containing both core generators and just one of $(4\ 5), (4\ 6)$ generate $\mathrm{Sym}(\{1,2,3,4,5\})$ and $\mathrm{Sym}(\{1,2,3,4,6\})$. Their intersection is $\mathrm{Sym}(Q)$, whereas the common marking subset generates only the order-eight core. This is a literal failure of the intersection property. The hypothesis is structural, not merely a convenience of the proof.

Let $G = \mathrm{Sym}(\Omega)$, $|\Omega| = n \geq 9$, have a C-group marking $\rho_0, \ldots, \rho_{n-3}$. In the permutation graph each transposition factor of ρ_i is an edge labelled i. Edges of one label form a matching. By the transitive-parabolic exclusion, every $G_i = \langle \rho_j : j \neq i \rangle$ is intransitive. For each i choose one i-edge joining distinct G_i-orbits. Such an edge exists, since otherwise ρ_i would preserve every G_i-orbit and G would be intransitive. Let F consist of these $n-2$ selected edges. The fracture-forest method is due to Fernandes, Leemans and Mixer; the following proof makes its complete use explicit.

4 The possible extra edges

Lemma 3. *The selected graph F is a forest with two components. Every extra edge joins its two components. Two extra edges intersect; if they have different endpoint sets, they and two selected edges form an alternating square. In that case they are the only extra edges. Consequently the permutation graph is of one of three forms:*

1. *a tree with exactly one edge label occurring twice;*
2. *a tree with a nonempty set of parallel extra edges at one place, every repeated label occurring exactly twice;*
3. *a single alternating square, with all other edges forming attached trees and having pairwise distinct labels absent from the square.*

In the second case there are at least two extra edges; in all cases every generator is a transposition or a product of two transpositions.

Proof. A cycle in F would express the endpoints of any selected i-edge as joined by a path using no i-edge, contradicting their distinct G_i-orbits. Thus F is a forest, and its $n-2$ edges give two components. If an extra i-edge had both endpoints in one component, its selected-tree path would have to contain no edge labelled $j \neq i$: otherwise the remaining cycle would join the selected j-edge's endpoints using

labels different from j. But the only path consisting solely of the single selected i-edge is that edge itself, which is not an extra edge. Hence every extra edge crosses between components.

Take two extra edges e_i, e_j. If they are disjoint, their endpoints are joined within the two components by nonempty selected paths. The cycle formed by these paths and the extra edges can use no selected label outside $\{i, j\}$. If $i = j$, the two nonempty paths already require more than the one selected i-edge. If $i \neq j$, each path has length at least two: a one-edge path between an endpoint of e_i and an endpoint of e_j can have neither label, since edges of a label form a matching. Again there are too many selected edges of the allowed labels. Thus the extra edges intersect, and in particular there cannot be two extra edges of one label.

If $e_i = \{a, b\}$ and $e_j = \{a, d\}$ with $b \neq d$, the selected path from b to d uses only i, j. The matching condition forces exactly two edges, labelled j, i in that order. This is the asserted square. There cannot be a third extra edge. All crossing edges are bipartite between the two components, so pairwise intersecting ones have a common endpoint. A third edge parallel to either existing one would force the same selected path to have two different final labels. A third with another endpoint would give three vertices at pairwise distance two in a tree. Their paths share one central vertex, but the edge at a fixed outer vertex would again have two required labels.

If there is only one extra edge, the resulting simple graph is a tree. If all extra edges are parallel, adjoining their common underlying edge to F also gives a tree. Otherwise the unique square just described is the only cycle and its two labels occur exactly twice. This proves all claims. □

5 Elementary splitting identities

All products below are ordinary permutations; disjoint factors commute.

Lemma 4. *Let τ and σ be disjoint transpositions.*

1. *If a transposition e meets exactly one of their supports in one point, then $\langle \tau\sigma, e\rangle$ contains both τ and σ.*
2. *For five distinct letters a, b, u, v, w, the three involutions $\tau(vu), \tau(vw), (va)$, where $\tau = (ab)$, generate the full symmetric group on those letters.*
3. *The subgroup $\langle \tau(vu), (va)\rangle$ does not contain τ.*

Proof. For (1), the product of the meeting pair is a three-cycle, so cubing $(\tau\sigma)e$ yields the other transposition; multiply back by $\tau\sigma$ to obtain both. For (2), the product of the first two involutions is a three-cycle on u, v, w. Conjugating (va) by its powers supplies (ua) and (wa). These generate the symmetric group on a, u, v, w, including (vu), and hence also τ; adding (ab) then gives the symmetric group on all five letters. For (3), both listed generators preserve the unordered partition $\{\{u, b\}, \{v, a\}\}$, whereas τ does not. □

Lemma 5. *Suppose all repeated labels have the same transposition factor $\tau = (45)$. Neither of the following marked five-letter configurations can be extended by an edge from its support to a sixth letter while preserving the intersection property:*

	r_0	r_1	r_2
path	$(12)\tau$	$(23)\tau$	(34)
star	$(13)\tau$	$(23)\tau$	(34).

(1)

The extending mark may be $(v6)$ for $1 \leq v \leq 5$, or $\tau(v6)$ for $1 \leq v \leq 3$.

Proof. In the path configuration, $\tau \in \langle r_0, r_2\rangle$ by Lemma 4(1). If the new mark $e = (v6)$ is untwisted, then τ also belongs to $\langle r_0, e\rangle$ for $v \in \{1, 2, 4, 5\}$, or to $\langle r_1, e\rangle$ for $v = 3$. The common distinguished

support is respectively $\{r_0\}$ or empty, neither of which generates τ. If $e = \tau(v6)$ with $v = 1$ or two, then $\tau \in \langle r_2, e\rangle$ by the same splitting identity; the common support with $\{r_0, r_2\}$ is only r_2. For $e = \tau(36)$, Lemma 4(2) puts τ in $\langle r_1, r_2, e\rangle$, again with common support only r_2.

In the star configuration, Lemma 4(2) says $\langle r_0, r_1, r_2\rangle = S_5$, which contains τ. Every untwisted $e = (v6)$ meets the support of at least one of r_0, r_1, and Lemma 4(1) puts τ in that generator together with e. The common support contains only that double transposition. If $e = \tau(v6)$ with $v = 1$ or two, then $\tau \in \langle r_2, e\rangle$ as above. For $e = \tau(36)$, Lemma 4(2) gives $\tau \in \langle r_0, r_2, e\rangle$. The common support is $\{r_0, r_2\}$, and Lemma 4(3) excludes τ from its subgroup. Every permitted extension therefore contradicts IP. □

6 Eliminating parallel edges and long repetitions

In the parallel or single-extra case of Lemma 3, write T for the underlying tree, obtained by adding the common extra edge $\tau = \{a, b\}$ to F. Each repeated mark is $\tau\sigma_i$; all other marks are single transpositions.

Lemma 6. *The two transpositions in every repeated mark are separated in T by exactly one edge.*

Proof. They are disjoint, so their distance is at least one. Suppose the path between σ_i and τ has at least two intervening edges. Let l be the first and j the last. Since j meets τ, its mark is a single transposition. If l is also single, each of $\langle \rho_i, \rho_l\rangle$ and $\langle \rho_i, \rho_j\rangle$ contains τ by splitting, whereas their common distinguished subgroup $\langle \rho_i\rangle$ does not. If l has repeated label, its mark is $\tau\sigma_l$, with $\sigma_l = l$.

When there are at least three intervening edges, j is disjoint from both σ_i and σ_l. Then each of $\langle \rho_i, \rho_j\rangle$ and $\langle \rho_l, \rho_j\rangle$ contains the symmetric group on the three letters supporting τ and j. Their intersection is larger than $\langle \rho_j\rangle$, contrary to IP. With exactly two intervening edges the local configuration is the path in (1), up to relabelling. Since $n \geq 9 > 5$ and T is connected, there is an edge from this five-letter subtree to its complement. Its label is new and has one of the eight forms in Lemma 5: an existing double label has already used both of its matching edges, and the existing single label meets τ and cannot be repeated. That lemma again contradicts IP. □

Lemma 7. *There are no parallel edges in the permutation graph.*

Proof. Suppose two distinct repeated marks are $\rho_i = \tau\sigma_i$ and $\rho_j = \tau\sigma_j$. By Lemma 6, each selected edge σ_i, σ_j has a single intervening connector to τ. These connectors are single transpositions, since they meet τ. If the connectors are distinct, the two selected edges are disjoint: otherwise the underlying graph would have a cycle. Let the connectors be k, l for σ_i, σ_j, respectively. The connector k meets τ and is disjoint from σ_j, while l is disjoint from σ_i. Splitting therefore puts τ in both $\langle \rho_j, \rho_k\rangle$ and $\langle \rho_i, \rho_l\rangle$. Their distinguished supports are disjoint, contradicting IP.

If the connectors coincide, the local configuration is the star in (1). Again connectedness and $n > 5$ supply an edge to a sixth letter, and Lemma 5 excludes every permitted label. □

7 The complete core classification

Theorem 8. *Every rank-$n-2$ C-group marking of S_n, $n \geq 9$, is, up to relabelling letters and generators, one of the three constructions:*

1. *core* $(23), (12)(34)$*, with a tree attached at the core vertex* 4*;*
2. *the same core, with a tree attached at vertex* 3*;*
3. *core* $(12)(34), (13)(24)$*, with a tree attached at vertex* 4*.*

In each case the core vertex is a leaf of the attached tree, its first edge is a single transposition, and every tree edge has a different new label. There are no other cases.

Proof. By the preceding lemmas the graph is either a simple tree with one repeated label at distance one, or a unique alternating square. The first gives a four-letter core consisting of two disjoint transpositions with a single connector; it is the first listed core, whose marked group is the transitive dihedral group D_8. The square gives the transitive Klein four core in the third construction.

For any transitive group $H \leq \mathrm{Sym}(Q)$ and a new letter x, the group $\langle H, (vx)\rangle$ is $\mathrm{Sym}(Q \cup \{x\})$: conjugation by H gives all star transpositions (qx), $q \in Q$. If two distinct edges leave one of our four-letter cores, with respective new endpoints x, y, the two groups generated by the core and one edge therefore intersect in $\mathrm{Sym}(Q)$. IP would require their intersection to be the core group, which is only D_8 or V_4. This is impossible. Hence exactly one edge leaves the core, and all remaining edges form one attached tree.

The marked dihedral core has two vertex orbits under permutations centralizing both of its distinguished generators: the two endpoints of the length-three path, and its two middle vertices. They give constructions A and B. To verify that these orbits are distinct, the single core transposition fixes exactly the two endpoints, whereas its support is the middle pair. Reversing the path interchanges the two vertices in each pair while preserving both marks. In the square case the Klein four group centralizes its own two marks and is transitive on the four vertices, giving only construction C. This proves exhaustion. The core-attachment theorem of [U] and its local verification above establish the full intersection property of every listed construction, and the flag-completion proof below establishes the regular-hypertope conclusion. □

Exact finite validation and source scope

The program `local_obstructions.py` independently enumerates subgroups of S_6 for all sixteen extensions in (1) and the two six-letter cases with distinct connectors. It emits explicit words for the common element τ and checks its exclusion from the subgroup on the common labels. Five deliberately corrupted certificates are rejected. These are validation examples: the splitting identities above prove all cases uniformly, so no finite enumeration is a premise of Theorem 8. Completeness of the local cases is established by the forest and distance arguments, not inferred from the program's case count.

The theorem recovers the necessity assertion of Fernandes–Leemans, Theorem 1.2, in the 2018 accepted manuscript, together with the intransitive-parabolic reduction proved separately. The fracture method and the three historical families are theirs. The module and simple-kernel bounds, explicit local splitting proof and decorated-coset flag completion are the present proof organization; no historical conjecture solution or priority claim follows from this recovery.

8 The objects of the coset system

Use the three rooted cores A, B and C and the notation of the attachment theorem. Thus $Q = \{1, 2, 3, 4\}$, $H = \langle h_0, h_1\rangle$, and a tree T meets Q at its leaf d. Its unique edge at d is denoted b. The group of the complete marking is $G = \mathrm{Sym}(\Omega)$. For each generator i, let G_i be the subgroup generated by all the other generators. Elements of type i are left cosets of G_i; incidence means nonempty intersection. We work with images of the following objects under permutations of Ω.

For an edge e of T, let S_e be the component of $T - e$ not containing d. For $e \neq b$, the attachment formula gives

$$G_e = \mathrm{Sym}(S_e) \times \mathrm{Sym}(\Omega \setminus S_e).$$

Thus an object of type e is a subset $X_e \subset \Omega$ of size $|S_e|$. The sets S_e form a laminar family: two of them are disjoint or one contains the other. Their images have the same pairwise incidence rule. Indeed, a permutation carries two prescribed subsets to two given subsets if and only if their four Boolean regions have the prescribed sizes. In the nested or disjoint cases this is precisely the stated containment or disjointness relation, together with the fixed individual sizes.

The bridge stabilizer is $G_b = H \times \mathrm{Sym}(\Omega \setminus Q)$. A bridge object is a four-set Q' together with the following decoration. In A and B it is a partition M of Q' into two unordered pairs, the image of $\{\{1,4\},\{2,3\}\}$. Its stabilizer on Q' is $D_8 = H$. In C it is an ordered list of the three perfect matchings on Q', the image of a fixed order of those on Q. The action of S_4 on these three matchings is onto S_3 and has kernel $\{1, (12)(34), (13)(24), (14)(23)\} = H$. This describes the bridge cosets exactly, including their full multiplicity.

The two remaining objects, of types 0 and 1, are as follows. Every displayed tuple has ordered entries; a displayed pair such as $\{a, b\}$ is unordered.

	type 0	type 1
A	$P = \{1,2\}$	$(p, R) = (1, \{2,3\})$
B	$P = \{1,2\}$	$(p, q) = (1, 4)$
C	$P = \{1,3\}$	$R = \{1,2\}$

These are exact coset coordinates, because the attachment formula gives their stabilizers as G_0 and G_1. In A, $p \notin R$; in B, $p \neq q$; in C each object is a two-subset. All their displayed points will be called their support.

Two such core objects are incident precisely under the following rules:

$$\begin{array}{c|l} A & p \in P, \quad |P \cap R| = 1, \quad p \notin R; \\ B & p \in P, \quad q \notin P; \\ C & |P \cap R| = 1. \end{array} \tag{2}$$

For example, in A these conditions identify the images of 1, 2 and 3 as p, the unique point of $P \cap R$, and the other point of R. The fourth core point may be any unused point. The other two assertions follow in the same way. In each case a compatible pair has support of size three; a single core object has support of size at most three.

A core object is incident with an ordinary tree object X_e precisely when its support is disjoint from X_e. Its stabilizer has one large block, containing the whole tail; that block is the complement of its support. Necessity follows by taking images of the standard objects. Conversely, disjointness lets one map the displayed core points first, then map S_e to X_e, and extend these disjoint partial bijections. The prescribed cardinalities guarantee enough remaining points. Similarly a bridge object (Q', δ) is incident with X_e precisely when $Q' \cap X_e = \varnothing$. These arguments apply also when the support has several distinguished blocks: the internal labels are fixed by the core-object choice and place no condition on the tail.

9 The finite decoration check

Lemma 9. *A flag consisting of a bridge object and any subset of the two core types has a representative bijection $Q \longrightarrow Q'$.*

Proof. Incidence with the bridge puts each support inside Q' and imposes its orbit under H. It remains to check simultaneous realization when both core types occur.

For A, a type-0 pair P is a transversal of the matching M. A type-1 object consists of one pair R of M and a point p of the other pair. By (2), p is the point of P outside R, and $P \cap R$ has one point. Map 1 to p, 2 to that intersection point, 3 to the other point of R, and 4 to the remaining point. This maps the standard matching and both standard core objects as required.

For B, incidence with the matching says that P is a transversal and (p, q) is an ordering of one of its pairs. Rule (2) puts p in P and q outside P. Map 1 to p, 4 to q, 2 to the other point of P, and 3 to the remaining point. Again the three objects are simultaneously realized.

For C, P is one pair of the matching assigned to type 0 and R is one pair of the different matching assigned to type 1. Distinct perfect matchings on a four-set have every pair meeting every pair in one

point. Map 1 to $P \cap R$, 2 to the other point of R, 3 to the other point of P, and 4 to the remaining point. The first two ordered matchings are correct; the third is then determined. This is a representative of the given bridge decoration as well.

If only one core type occurs, any incident representative already provides the required bijection. The empty case is immediate. □

10 Completion of a tree flag

We give explicitly the weighted-tree fact used here. Regard the root d as a vertex of weight four, occupied by the whole set Q, and every other tree vertex as having weight one. For selected edges $e \neq b$, suppose subsets X_e of the prescribed sizes have all their prescribed pairwise nesting and disjointness relations. For a selected edge e, let its selected children mean the maximal selected descendants below it. The differences

$$X_e \setminus \bigcup_{f \text{ a selected child of } e} X_f$$

have exactly the cardinalities of the analogous differences among the standard S_e. The root region

$$D = \Omega \setminus \bigcup_{e \text{ maximal selected}} X_e$$

has the cardinality of the analogous standard root region, and hence has at least four points. All these regions are disjoint. A bijection on a chosen four-subset of D can therefore be extended independently on each difference region to a permutation taking every S_e to X_e. This is the classical laminar flag-completion argument, with root weight four, written in full; no classification theorem is needed for it.

Theorem 10. *For each of the three cores, and every attached tree allowed by the attachment theorem, the coset system of the complete marking is flag-transitive. It is a thin, residually connected geometry, and the specified symmetric group acts regularly on its chambers.*

Proof. Take an arbitrary flag and collect its ordinary tree objects X_e. They have the nesting and disjointness relations above, so construct the root region D. Every core-object support is contained in D, because it is disjoint from every selected X_e.

If the flag contains a bridge object, its four-set Q' also lies in D. Lemma 9 gives a bijection $Q \to Q'$ realizing all the flag's core and bridge objects. If the bridge is absent, then there are at most two core objects. Pairwise incidence gives (2), and hence a partial bijection from at most three core points to their support. Extend it to a bijection $Q \to Q'$ for some four-subset Q' of D. This is possible because $|D| \geq 4$. With just one or no core object the same construction is simpler.

Now extend that bijection by the laminar completion just proved. The resulting permutation realizes every object in the flag simultaneously. It belongs to all of the corresponding cosets, so every flag extends to a chamber and every flag of fixed type is in the orbit of the standard flag. Thus the coset system is flag-transitive.

For completeness, the attachment theorem identifies every intersection of maximal parabolics with the subgroup generated by the remaining marks. The intersection of all maximal parabolics is trivial, so the chamber action is free. A panel stabilizer is the cyclic group generated by its omitted involution and has order two; hence panels have exactly two chambers and the geometry is thin. In any residue of rank at least two, the remaining maximal parabolics generate its group, since each remaining generator belongs to at least one of those parabolics. Indeed, for a coset system, the union of the subgroups specifying its types generates a group precisely when its incidence graph is connected: a word in those subgroups gives a path through successive cosets, and conversely such a path gives a word. Applying this to each residue proves residual connectedness. The chamber graph is connected as well, since the involutions generate G. A type-preserving automorphism fixing one chamber must fix its unique

adjacent chamber of each type, by thinness, and hence every chamber. It fixes every object, so the full type-preserving automorphism group is exactly G. □

11 Exact presentations on the three markings

Apply the presentation-transport lemma of [U] to the three attached cores. Let $\beta = (d\ x)$ be the bridge transposition and leave all tail transpositions unchanged. For A and B take the core tree with transpositions

$$a = (12), \quad b = (23), \quad c = (34).$$

Then $h_0 = b$, $h_1 = ac$, and the inverse words are

$$a = (h_1\beta)^3, \qquad b = h_0, \qquad c = (h_1\beta)^3 h_1. \tag{3}$$

These identities hold for either $d = 4$ or $d = 3$: the bridge touches one factor of the double transposition, and cubing its product with h_1 isolates the other factor (12).

For C take the core tree with

$$a = (12), \quad b = (13), \quad c = (34).$$

The forward and inverse words are

$$h_0 = ac, \qquad h_1 = babcba, \qquad a = (h_0\beta)^3, \quad b = (h_1\beta)^3, \quad c = (h_0\beta)^3 h_0. \tag{4}$$

Indeed $abcba = (24)$, so $babcba = (13)(24)$; the cube identities isolate (12) and (13) respectively. In each case the three core edges and the full tail form a tree on Ω. Substituting (3) or (4) into the tree-presentation theorem of [U] and adding the two forward marking identities gives, by the presentation-transport lemma of [U], a complete finite presentation on the original $|\Omega| - 2$ generators. This construction handles arbitrary branching and specifies every relator by an explicit rule.

11.1 The Coxeter diagrams and the printed local words

Every pair order is determined by the following table, together with incidence of edges in the tail:

	$\lvert h_0h_1\rvert$	$\lvert h_0\beta\rvert$	$\lvert h_1\beta\rvert$
A	4	2	6
B	4	3	6
C	2	6	6

All other tail marks commute with both core marks. Two tail marks have product order three if their tree edges meet, and order two otherwise. In particular the bridge has order-three products with *every* edge at its other endpoint. To check the table, the two D8 core products are four-cycles and the V4 core generators commute. A transposition meeting just one factor of a double transposition has a disjoint three-cycle and transposition as product, of order six. The two remaining entries for $h_0\beta$ are the disjoint or one-point-intersection cases for transpositions. This proves the complete Coxeter-diagram description for every tail.

Let t be any tail edge adjacent to the bridge at x. In A and B write $a = (h_1\beta)^3 = (12)$. Then $|ah_0| = 3$ and $[a, t] = 1$. In A, $h_1\beta h_1 = (3x)$, so $(h_0h_1\beta h_1)^3 = 1$. In B, $h_1h_0h_1 = (14)$ is disjoint from $\beta = (3x)$, so $(h_1h_0h_1\beta)^2 = 1$. In C write $a = (h_0\beta)^3 = (12)$ and $b = (h_1\beta)^3 = (13)$. Then $|ab| = 3$, $abh_1 = (12)(24)$ has order three, and both a and b commute with t. These are precisely the local extra-word identities in [FL, Corollary 6.5(5)], in the displayed coordinate conventions. Every triangle of tail edges also satisfies the triangle relation of [U]. Thus the local diagram and word claims are retained wherever their stated markings apply; the following qualification concerns the uniform arbitrary-branching presentation.

12 A literal branching qualification

The source's accepted manuscript (institutional repository, `10773/23861`, physical pages 17–18, Corollary 6.5) states that ρ_2 commutes with all ρ_i for $i \geq 4$, while Theorem 1.2 permits arbitrary branching immediately after the bridge. These clauses cannot both hold literally. On nine letters, take

$$(23),\ (12)(34),\ (45),\ (56),\ (57),\ (68),\ (79). \tag{5}$$

The attachment theorem proves the full intersection property and identifies the generated group as S_9. The complete flag proof also applies. But $(45)(56)$ and $(45)(57)$ both have order three. Thus the bridge fails to commute with two tail generators, independently of how those tail labels are permuted. The double transposition and its unique order-six neighbor identify the bridge within this marked family, so a renumbering preserving the specified core cannot repair that assertion.

The corrected general presentation is obtained from the transport rule of [U] with (3)–(4). For a path tail it specializes to the familiar changes of marking used in the source. We do not assert, without a separate Tietze calculation, that every redundant transported relation follows from the source's printed shorter path list. Nor do we describe the false branching clause as a recovered theorem. It is an explicit correction within the intended general presentation result.

13 Boundary example and exact recovery scope

The restriction $n \geq 9$ in the rank-$n-2$ assertion cannot be removed unchanged. The accepted source displays the following six involutions on eight letters:

$$(12),\ (12)(34),\ (12)(78),\ (12)(56),\ (13)(68),\ (18)(36).$$

For the finite flag check we use the following elementary criterion. If H_i are subgroups, all coset flags have common representatives if and only if, for every anchor a and set J of other types, $\bigcap_{j\in J} H_a H_j = H_a(\bigcap_{j\in J} H_j)$. Indeed, inductively realize the flag on J and translate its common representative to the identity. Its objects are then the subgroups H_j. A right coset $H_a x$ is incident with each of them exactly when x belongs to the intersection on the left; it meets their common intersection exactly when x belongs to the right side. This proves both necessity and the induction step. It is the product-set criterion recalled in [FL, Theorem 2.5], here with the coset convention reversed by inversion. The program represents $H_a H_j$ by the orbit of H_a under H_j on right cosets, so its test is exactly this criterion.

The complete auxiliary program `boundary_example.g` verifies that they generate S_8, that all $64^2 = 4096$ subset intersections satisfy the C-group condition, and that their coset system is flag-transitive. It checks every product-set condition using all 156 choices of anchor and at least two remaining types. The maximal parabolic obtained by omitting the first generator is transitive of order 1344. This bounded example is a computational recovery of the source's stated boundary example, not a premise in either uniform classification proof. The same program verifies the two order-three bridge products in the branching correction above; its all-subset intersection and geometry assertions already follow from the uniform written theorems.

The three families, their geometric conclusions and the historical fracture method belong to Fernandes–Leemans and the prior work they cite, including Fernandes–Leemans–Mixer. The maximum independent length and omission alternatives retain Whiston through the exact Cameron–Cara interfaces listed in [U]. The new proof organization uses the repeated normal-module argument, a simple-factor chain bound, one finite attachment interface and the explicit decorated-flag completion. These arguments are written in full; no earlier combined manuscript is an unpriced mathematical premise.

The source's auxiliary table of primitive C-ranks is not recovered as a table of exact ranks. The required program proves only the complete primitive bounds and exact finite group facts needed by

our converse. We do not infer that a string C-rank equals an unrestricted C-rank. The unnumbered statement about three times a number of leaves is also not made a counting contract: after forgetting generator labels one must specify whether leaves are distinguished individually or only up to automorphisms of the tail. The classification of all markings up to relabelling is proved, but a separate numerical convention for that ancillary sentence is left explicit rather than assumed. The background results on general incidence systems and on arbitrary hypertopes are used only in the elementary coset form proved here; this paper does not claim to replace every historical background assertion or every maximal-parabolic case description verbatim.

The entire common paper is a dependency of this application. The primitive catalogue program is a required finite input for its necessity proof. The four-point identities and all local splitting cases have complete written proofs; their two full programs are validation. The boundary example program is required only for that additional finite source example and is not required for the five principal results. All four programs are printed without pruning. Operational drivers and the eight-declaration Lean certificate are separate evidence. The certificate covers the generic normalization leg; it does not formalize this classification, the catalogue, the finite-core theorem or flag completion. No compression ratio, additional source admission or question credit is inferred before independent review of the complete packet.

References

[FL] M. E. Fernandes and D. Leemans, C-groups of high rank for the symmetric groups, *J. Algebra* **508** (2018), 196–218, doi:10.1016/j.jalgebra.2018.04.031. Exact source used: accepted manuscript, University of Aveiro repository item 10773/23861, 21 physical pages including publisher cover. The original arXiv:1510.08990v1 is retained separately for edition comparison.

[CC] P. J. Cameron and P. Cara, Independent generating sets and geometries for symmetric groups, *J. Algebra* **258** (2002), 641–650.

[U] Human Compression project, *Irredundant generation and finite cores on trees*, companion common theory in this packet; full proofs and exact retained input statement, author edition awaiting independent review.